Signals and Systems Course # EE 2235
Lecture #3 Topic: Basic Operations on Signals

- Operations performed on dependent variables
  - Amplitude scaling
  - Addition
  - Multiplication
  - Differentiation
  - Integration

- Operations performed on the independent variables
  - Time scaling
  - Reflection
  - Time shifting

(1) Operations performed on dependent variables

(i) Amplitude or Magnitude Scaling: Example – electronic amplifier

CT signal: $y(t) = cx(t)$, where c is the scaling factor; voltage = resistor*current

DT signal: $y[n] = cx[n]$

(ii) Addition: Example - audio mixer, which combines music and voice signals

CT signal: $y(t) = x_1(t) + x_2(t)$

DT signal: $y[n] = x_1[n] + x_2[n]$

(iii) Multiplication: Example – AM radio signal

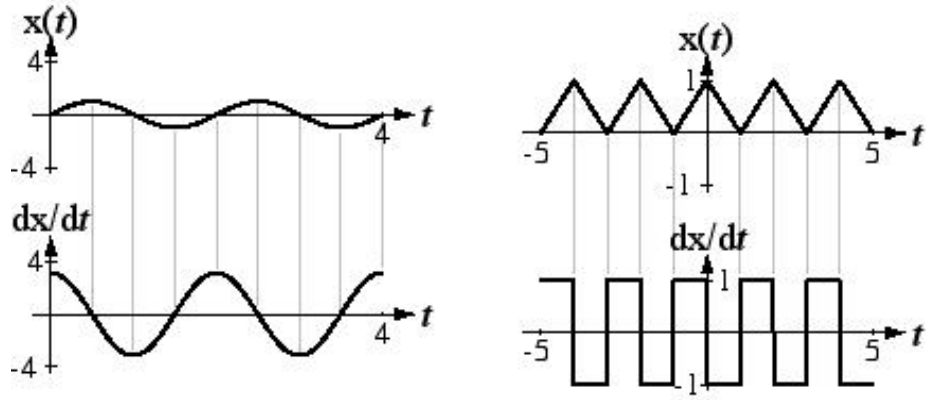
CT signal: $y(t) = x_1(t)x_2(t)$;

$x_1(t)$ = audio signal + a dc component, $x_2(t)$=sinusoidal signal (carrier)

DT signal: $y[n] = x_1[n]x_2[n]$

(iv) Differentiation: Inductor performs differentiation

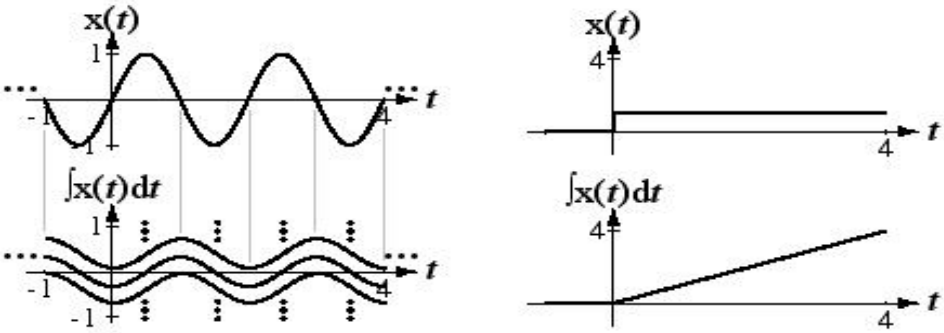
CT signal: $y(t) = \frac{d}{dt}x(t)$, $v(t) = L\frac{d}{dt}i(t)$



(v) Integration: A capacitor performs integration

CT signal: $y(t) = \int_{-\infty}^{t} x(\tau)d\tau$, where τ is the integration variable

Voltage develops across the capacitor, $v(t) = \frac{1}{C}\int_{-\infty}^{t} i(\tau)d\tau$



(2) Operations performed on independent variables

(i) Time scaling: output is obtained by scaling the independent variable

CT signal: $y(t) = x(at)$

DT signal: $y[n] = x[kn]$, $k>0$

If $a>1$, the signal y(t) is a compressed version of x(t).
If $0<a<1$, the signal y(t) is an extended (stretched) version of x(t)

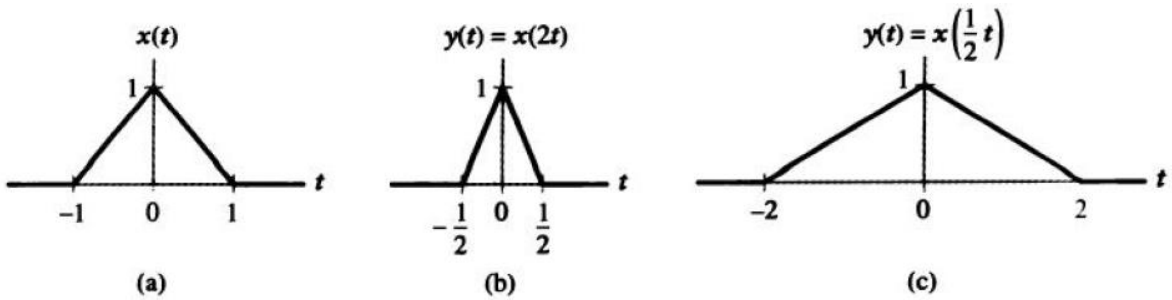


FIGURE 1.20 Time-scaling operation: (a) continuous-time signal $x(t)$, (b) version of $x(t)$ compressed by a factor of 2, and (c) version of $x(t)$ expanded by a factor of 2.

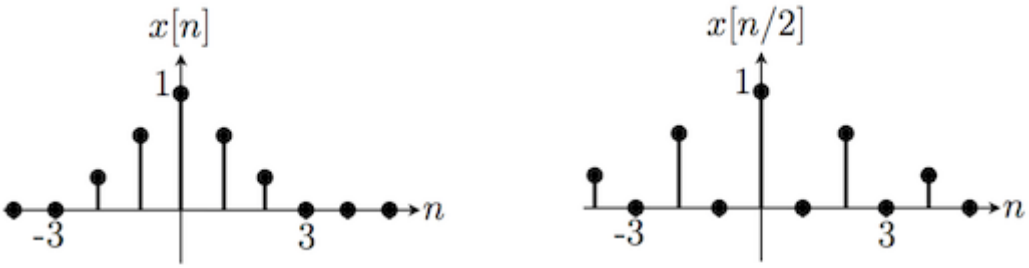


FIGURE 1.21 Effect of time scaling on a discrete-time signal: (a) discrete-time signal $x[n]$ and

(ii) Reflection:

CT signal: $y(t) = x(-t)$

DT signal: $y[n] = x[-n]$

The signal y(t) represents a reflected version of x(t) about the amplitude axis

Two cases:

Even signals: ***x*(-t) = *x*(t) for all t**; an even signal is same as the reflected version

Odd signals: ***x*(-t) = – *x*(t) for all t**; an odd signal is the negative of its reflected version

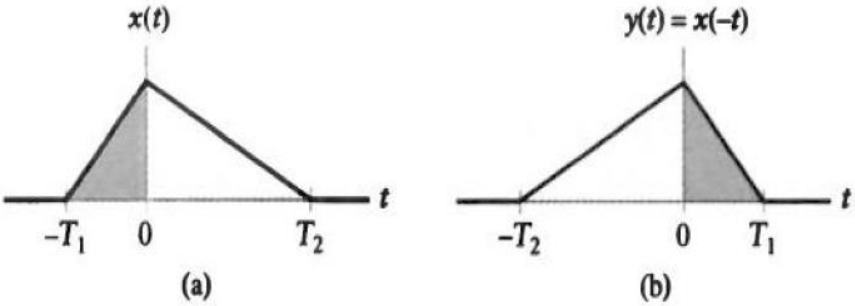


FIGURE 1.22 Operation of reflection: (a) continuous-time signal $x(t)$ and (b) reflected version of $x(t)$ about the origin.

Here, $x(t) = 0$ for $t < -T_1$ and $t > T_2$

$y(t) = 0$ for $t > T_1$ and $t < -T_2$

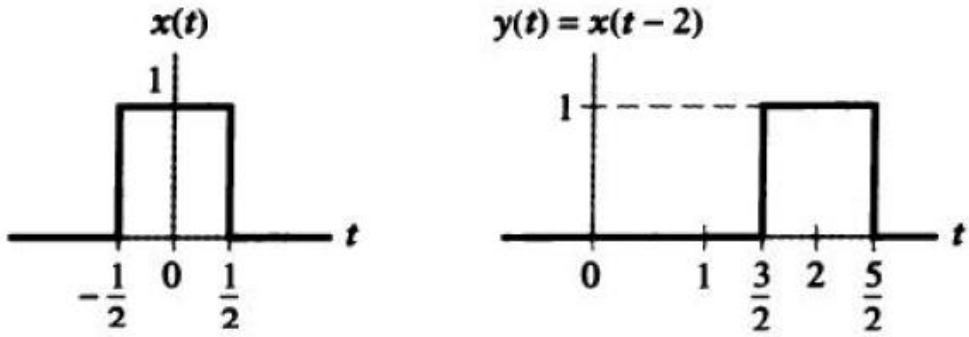
(vii) Time-shifting: Time-shifted version of *x*(t) is defined by

CT signal: $y(t) = x(t - t_0)$, where t_0 is the time shift

If $t_0 > 0$, the waveform representing $x(t)$ is shifted intact to the right, relative to the time axis

If $t_0 < 0$, it is shifted to the left

DT signal: $y[n] = x[n - m]$, where the shift m must be a positive or negative integer

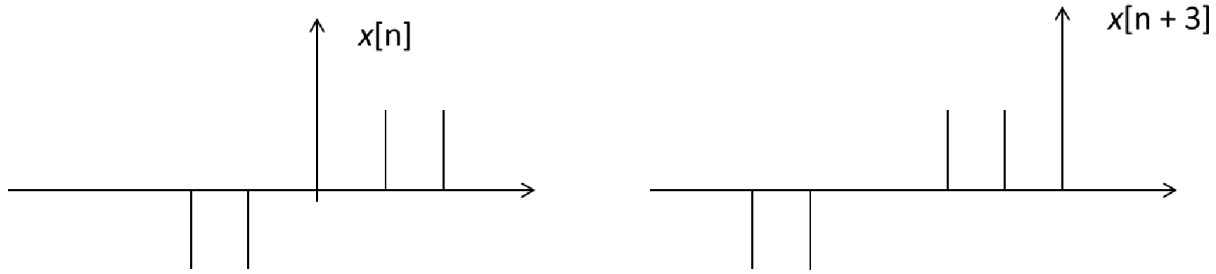


▶ Problem 1.13 The discrete-time signal

$$x[n] = \begin{cases} 1, & n = 1, 2 \\ -1, & n = -1, -2 \\ 0, & n = 0 \text{ and } |n| > 2 \end{cases}.$$

Find the time-shifted signal $y[n] = x[n + 3]$.

Answer: After drawing the figure, it can be easily shown that



$$y[n] = \begin{cases} 1, & n = -1, -2 \\ -1, & n = -4, -5 \\ 0, & n = -3, n < -5, \text{and } n > -1 \end{cases}.$$

Multiple transformations: $g(t) \to Ag\left(\frac{t - t_0}{a}\right)$**: Amplitude scaling, time scaling, and time shifting can be applied simultaneously. How?**

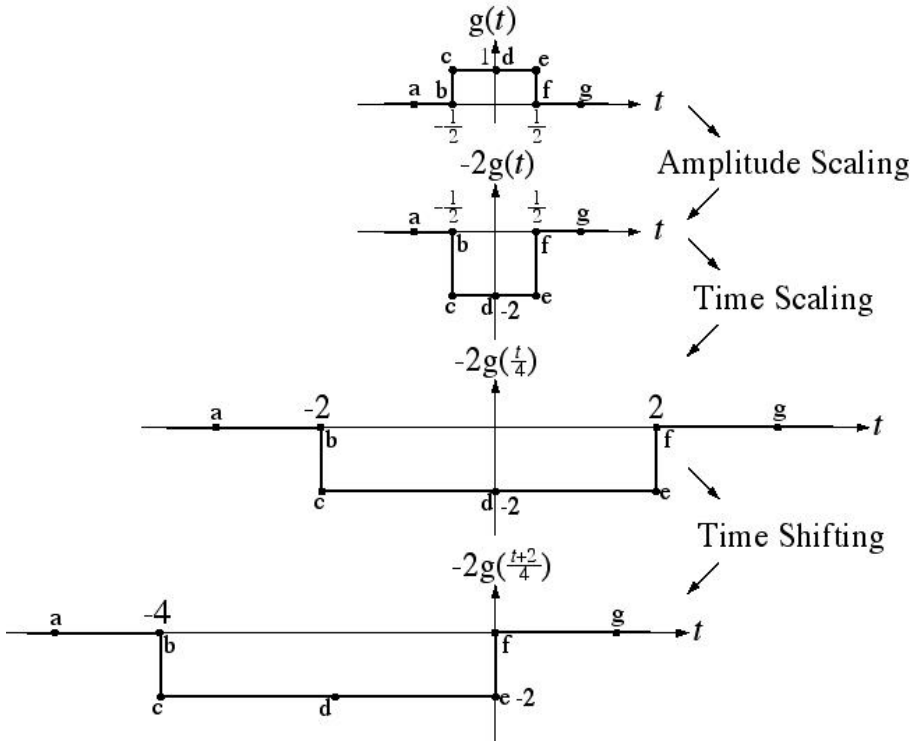
Answer: Multiple transformations can be done in steps

$$g(t) \xrightarrow[\text{scaling}, A]{\text{amplitude}} Ag(t) \xrightarrow{t \to t/a} Ag\left(\frac{t}{a}\right) \xrightarrow{t \to t - t_0} Ag\left(\frac{t - t_0}{a}\right)$$

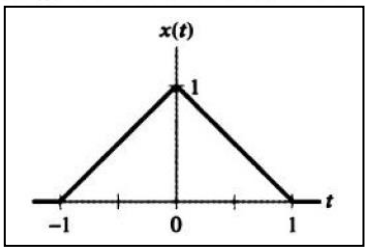
The order of the changes is important. For example, if we exchange the order of the time-scaling and time-shifting operations, we get:

$$g(t) \xrightarrow[\text{scaling},A]{\text{amplitude}} Ag(t) \xrightarrow{t \to t-t_0} Ag(t-t_0) \xrightarrow{t \to t/a} Ag\left(\frac{t}{a}-t_0\right) \neq Ag\left(\frac{t-t_0}{a}\right)$$

#A sequence of amplitude scaling, time scaling, and time shifting: $g(t) \to Ag\left(\frac{t-t_0}{a}\right)$

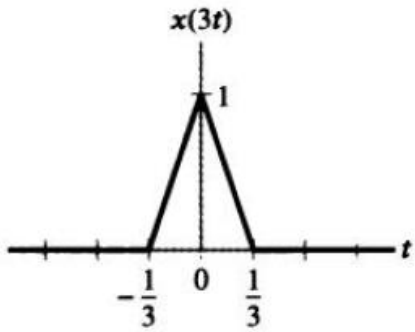


► **Problem 1.14** A triangular pulse signal $x(t)$ is depicted in Fig. 1.26. Sketch each of the following signals derived from $x(t)$:

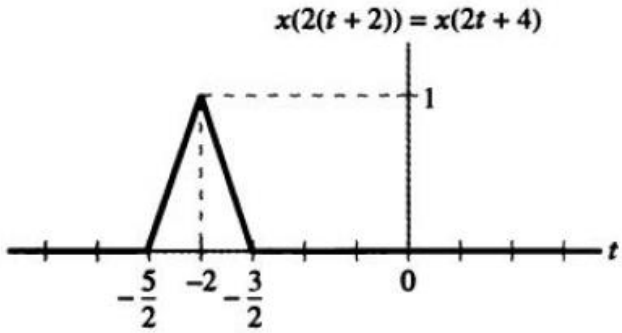


(a) $x(3t)$
(b) $x(3t+2)$
(c) $x(-2t-1)$
(d) $x(2(t+2))$
(e) $x(2(t-2))$
(f) $x(3t)+x(3t+2)$

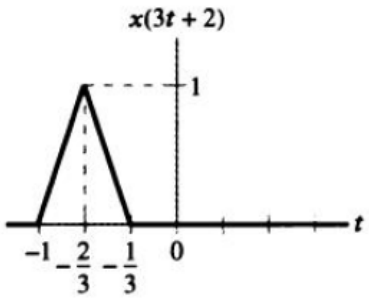
Answer:



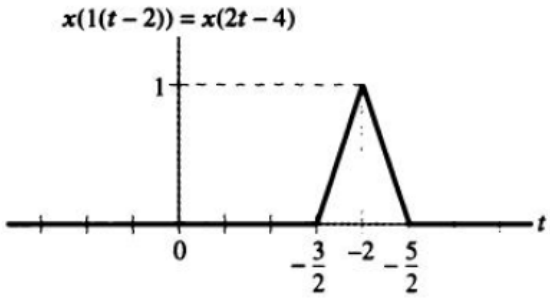
(a)



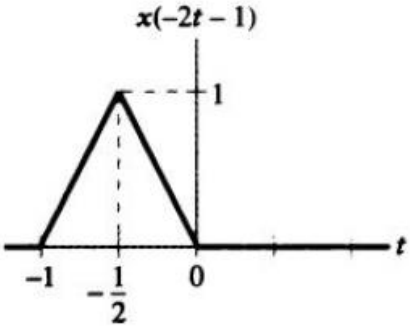
(d)



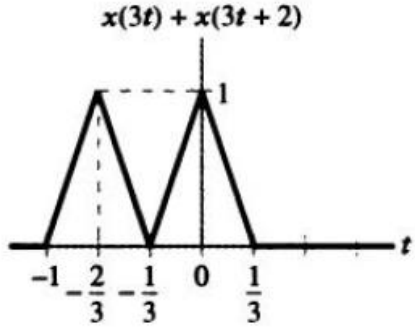
(b)



(e)



(c)



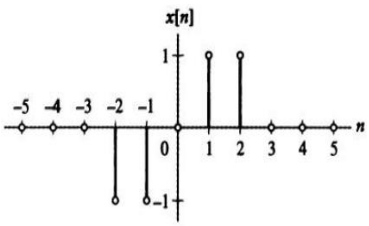
(f)

EXAMPLE 1.6 PRECEDENCE RULE FOR DISCRETE-TIME SIGNAL A discrete-time signal is defined by

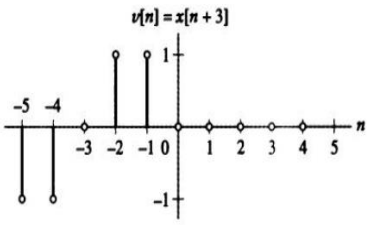
$$x[n] = \begin{cases} 1, & n = 1, 2 \\ -1, & n = -1, -2 \\ 0, & n = 0 \text{ and } |n| > 2 \end{cases}.$$

Find $y[n] = x[2n + 3]$.

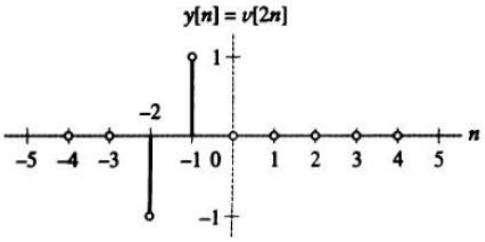
Answer:



(a)



(b)



(c)